

Naive Bayes Classifier

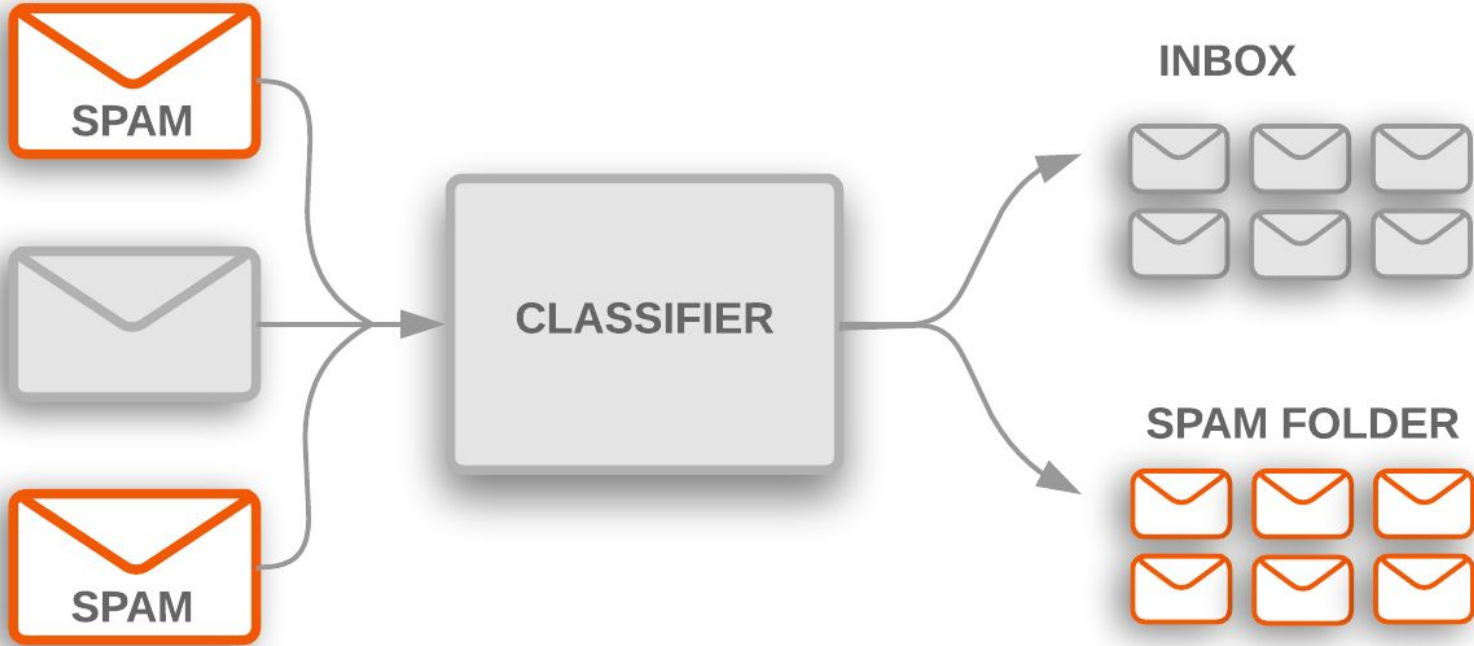


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Things We'd Like to Do

- Spam Classification
 - Given an email, predict whether it is spam or not
- Medical Diagnosis
 - Given a list of symptoms, predict whether a patient has disease X or not
- Weather
 - Based on temperature, humidity, etc... predict if it will rain tomorrow

Practical Example: Spam or not



Practical Example: Spam or not

Text Data

Bag of words

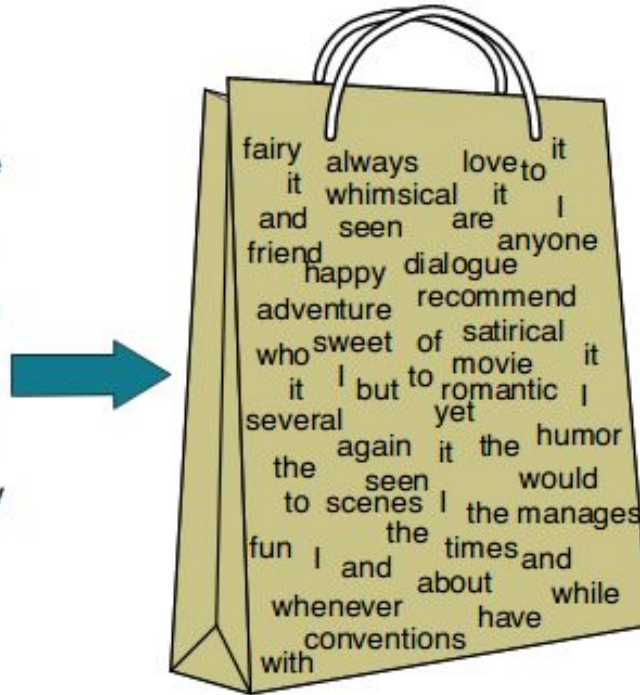
[
 'small dog',
 'cute cute cat',
 'cute dog'
]



	cat		cute		dog		small	
	0		0		1		1	
	1		2		0		0	
	0		1		1		0	

Practical Example: Spam or not

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet!



it	6
I	5
the	4
to	3
and	3
seen	2
yet	1
would	1
whimsical	1
times	1
sweet	1
satirical	1
adventure	1
genre	1
fairy	1
humor	1
have	1
great	1
...	...

Naïve Bayes Classifier

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$



Thomas Bayes
1702 - 1761

The Bayes Classifier

- A good strategy is to predict:

$$\arg \max_Y P(Y|X_1, \dots, X_n)$$

- (for example: what is the probability that it is spam given n dimension Bag of words features?)
-
- So ... How do we compute that?

- Use Bayes Rule!

$$P(Y|X_1, \dots, X_n) = \frac{\overset{\text{Likelihood}}{P(X_1, \dots, X_n|Y)} \overset{\text{Prior}}{P(Y)}}{\underset{\text{Normalization Constant}}{P(X_1, \dots, X_n)}}$$

- Why did this help? Well, we think that we might be able to specify how features are “generated” by the class label

- Let's expand this for our SPAM Classification task (Spam:5 and Non-spam:6):

$$P(Y = 5|X_1, \dots, X_n) = \frac{P(X_1, \dots, X_n|Y = 5)P(Y = 5)}{P(X_1, \dots, X_n|Y = 5)P(Y = 5) + P(X_1, \dots, X_n|Y = 6)P(Y = 6)}$$
$$P(Y = 6|X_1, \dots, X_n) = \frac{P(X_1, \dots, X_n|Y = 6)P(Y = 6)}{P(X_1, \dots, X_n|Y = 5)P(Y = 5) + P(X_1, \dots, X_n|Y = 6)P(Y = 6)}$$

- To classify, we'll simply compute these two probabilities and predict based on which one is greater

Model Parameters

- The problem with explicitly modeling $P(X_1, \dots, X_n | Y)$ is that there are usually way too many parameters:
 - We'll run out of space
 - We'll run out of time
 - And we'll need tons of training data (which is usually not available)

The Naïve Bayes Model

- The *Naïve Bayes Assumption*: Assume that all features are independent **given the class label Y**
- Equationally speaking:

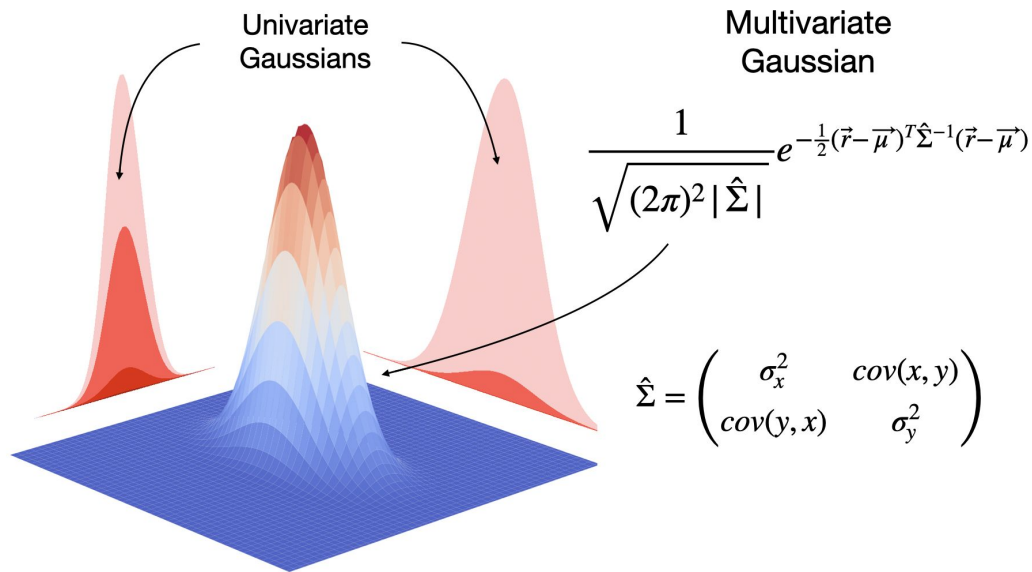
$$P(X_1, \dots, X_n | Y) = \prod_{i=1}^n P(X_i | Y)$$

- (We will discuss the validity of this assumption later)

The Naïve Bayes Model

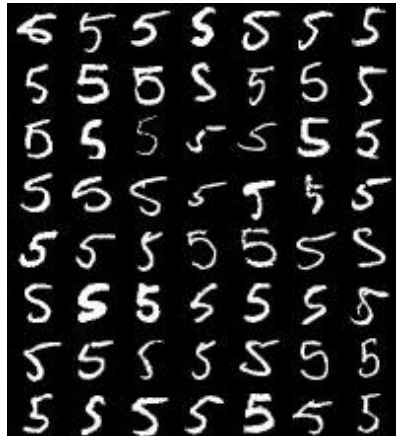
- The *Naïve Bayes Assumption*: Assume that all features are independent **given the class label Y**
- Equationally speaking:

$$P(X_1, \dots, X_n | Y) = \prod_{i=1}^n P(X_i | Y)$$



Naïve Bayes Training

- Now that we've decided to use a Naïve Bayes classifier, we need to train it with some data:



MNIST Training Data

Naïve Bayes Training

- Training in Naïve Bayes is **easy**:
 - Estimate $P(Y=v)$ as the fraction of records with $Y=v$

$$P(Y = v) = \frac{\text{Count}(Y = v)}{\# \text{ records}}$$

- Estimate $P(X_i=u | Y=v)$ as the fraction of records with $Y=v$ for which $X_i=u$

$$P(X_i = u | Y = v) = \frac{\text{Count}(X_i = u \wedge Y = v)}{\text{Count}(Y = v)}$$

- (This corresponds to Maximum Likelihood estimation of model parameters)

	Outlook	Temperature	Humidity	Windy	Play Golf
0	Rainy	Hot	High	False	No
1	Rainy	Hot	High	True	No
2	Overcast	Hot	High	False	Yes
3	Sunny	Mild	High	False	Yes
4	Sunny	Cool	Normal	False	Yes
5	Sunny	Cool	Normal	True	No
6	Overcast	Cool	Normal	True	Yes
7	Rainy	Mild	High	False	No
8	Rainy	Cool	Normal	False	Yes
9	Sunny	Mild	Normal	False	Yes
10	Rainy	Mild	Normal	True	Yes
11	Overcast	Mild	High	True	Yes
12	Overcast	Hot	Normal	False	Yes
13	Sunny	Mild	High	True	No

	Outlook	Temperature	Humidity	Windy	Play Golf
0	Rainy	Hot	High	False	No
1	Rainy	Hot	High	True	No
2	Overcast	Hot	High	False	Yes
3	Sunny	Mild	High	False	Yes
4	Sunny	Cool	Normal	False	Yes
5	Sunny	Cool	Normal	True	No
6	Overcast	Cool	Normal	True	Yes
7	Rainy	Mild	High	False	No
8	Rainy	Cool	Normal	False	Yes
9	Sunny	Mild	Normal	False	Yes
10	Rainy	Mild	Normal	True	Yes
11	Overcast	Mild	High	True	Yes
12	Overcast	Hot	Normal	False	Yes
13	Sunny	Mild	High	True	No

$$P(\text{Play Golf} = \text{Yes}) = 9/14$$

$$P(\text{Play Golf} = \text{No}) = 5/14$$

	Outlook	Temperature	Humidity	Windy	Play Golf
0	Rainy	Hot	High	False	No
1	Rainy	Hot	High	True	No
2	Overcast	Hot	High	False	Yes
3	Sunny	Mild	High	False	Yes
4	Sunny	Cool	Normal	False	Yes
5	Sunny	Cool	Normal	True	No
6	Overcast	Cool	Normal	True	Yes
7	Rainy	Mild	High	False	No
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10	Rainy	Mild	Normal	True	Yes
11	Overcast	Mild	High	True	Yes
12	Overcast	Hot	Normal	False	Yes
13	Sunny	Mild	High	True	No

Frequency Table

		Play Golf	
		Yes	No
Outlook	Sunny	3	2
	Overcast	4	0
	Rainy	2	3

		Play Golf	
		Yes	No
Humidity	High	3	4
	Normal	6	1

		Play Golf	
		Yes	No
Temp.	Hot	2	2
	Mild	4	2
	Cool	3	1

		Play Golf	
		Yes	No
Windy	False	6	2
	True	3	3

Likelihood Table

		Play Golf	
		Yes	No
Outlook	Sunny	3/9	2/5
	Overcast	4/9	0/5
	Rainy	2/9	3/5

		Play Golf	
		Yes	No
Humidity	High	3/9	4/5
	Normal	6/9	1/5

		Play Golf	
		Yes	No
Temp.	Hot	2/9	2/5
	Mild	4/9	2/5
	Cool	3/9	1/5

		Play Golf	
		Yes	No
Windy	False	6/9	2/5
	True	3/9	3/5

Probability of playing:

$$P(\text{Play} = \text{Yes} \mid \text{Weather} = \text{Overcast}, \text{Temp} = \text{Mild}) = P(\text{Weather} = \text{Overcast}, \text{Temp} = \text{Mild} \mid \text{Play} = \text{Yes})P(\text{Play} = \text{Yes}) \dots\dots\dots(1)$$

$$P(\text{Weather} = \text{Overcast}, \text{Temp} = \text{Mild} \mid \text{Play} = \text{Yes}) = P(\text{Overcast} \mid \text{Yes})P(\text{Mild} \mid \text{Yes}) \dots\dots\dots(2)$$

1. Calculate Prior Probabilities: $P(\text{Yes}) = 9/14 = 0.64$
2. Calculate Posterior Probabilities: $P(\text{Overcast} \mid \text{Yes}) = 4/9 = 0.44$ $P(\text{Mild} \mid \text{Yes}) = 4/9 = 0.44$
3. Put Posterior probabilities in equation (2) $P(\text{Weather} = \text{Overcast}, \text{Temp} = \text{Mild} \mid \text{Play} = \text{Yes}) = 0.44 * 0.44 = 0.1936$ (Higher)
4. Put Prior and Posterior probabilities in equation (1) $P(\text{Play} = \text{Yes} \mid \text{Weather} = \text{Overcast}, \text{Temp} = \text{Mild}) = 0.1936 * 0.64 = 0.124$

Probability of not playing:

$$P(\text{Play} = \text{No} \mid \text{Weather} = \text{Overcast}, \text{Temp} = \text{Mild}) = P(\text{Weather} = \text{Overcast}, \text{Temp} = \text{Mild} \mid \text{Play} = \text{No}) P(\text{Play} = \text{No}) \dots\dots\dots(3)$$

$$P(\text{Weather} = \text{Overcast}, \text{Temp} = \text{Mild} \mid \text{Play} = \text{No}) = P(\text{Weather} = \text{Overcast} \mid \text{Play} = \text{No}) P(\text{Temp} = \text{Mild} \mid \text{Play} = \text{No}) \dots\dots\dots(4)$$

1. Calculate Prior Probabilities: $P(\text{No}) = 5/14 = 0.36$
2. Calculate Posterior Probabilities: $P(\text{Weather} = \text{Overcast} \mid \text{Play} = \text{No}) = 0/9 = 0$ $P(\text{Temp} = \text{Mild} \mid \text{Play} = \text{No}) = 2/5 = 0.4$
3. Put posterior probabilities in equation (4) $P(\text{Weather} = \text{Overcast}, \text{Temp} = \text{Mild} \mid \text{Play} = \text{No}) = 0 * 0.4 = 0$
4. Put prior and posterior probabilities in equation (3) $P(\text{Play} = \text{No} \mid \text{Weather} = \text{Overcast}, \text{Temp} = \text{Mild}) = 0 * 0.36 = 0$

M-Estimate Approach

- In practice, some of these counts can be zero
- Fix this by adding “virtual” counts (for two class problem):

$$P(X_i = u|Y = v) = \frac{\text{Count}(X_i = u \wedge Y = v) + mp}{\text{Count}(Y = v) + m}$$

- P - Prior estimate of the probability
- m - Equivalent sample size

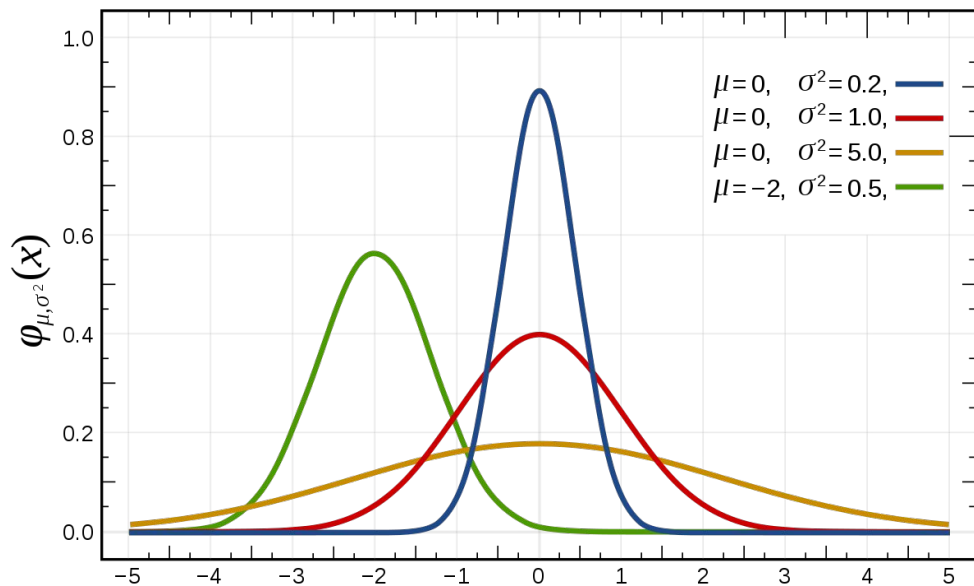
The Naive Bayes Classifier for Data Sets with Numerical Attribute Values

- One common practice to handle numerical attribute values is to assume normal distributions for numerical attributes.
- Let $x_1, x_2, \dots, x_n$ be the values of a numerical attribute in the training data set.

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\sigma = \frac{1}{n-1} \sum_{i=1}^n (x_i - \mu)^2$$

$$f(w) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(w-\mu)^2}{\sigma^2}}$$



- For examples,

$$f(\text{temperature} = 66 \mid \text{Yes}) = \frac{1}{\sqrt{2\pi}(6.2)} e^{-\frac{(66-73)^2}{2(6.2)^2}} = 0.0340$$

- Likelihood of Yes = $\frac{2}{9} \times 0.0340 \times 0.0221 \times \frac{3}{9} \times \frac{9}{14} = 0.000036$

- Likelihood of No = $\frac{3}{5} \times 0.0291 \times 0.038 \times \frac{3}{5} \times \frac{5}{14} = 0.000136$

Numerical Stability

- It is often the case that machine learning algorithms need to work with very small numbers
 - Imagine computing the probability of 2000 independent coin flips
 - Python thinks that $(.5)^{2000}=0$

Underflow Prevention

- Multiplying lots of probabilities

□ floating-point underflow.

- Recall: $\log(xy) = \log(x) + \log(y)$,

□ better to sum logs of probabilities rather than multiplying probabilities.

Underflow Prevention

- Class with highest final un-normalized log probability score is still the most probable.

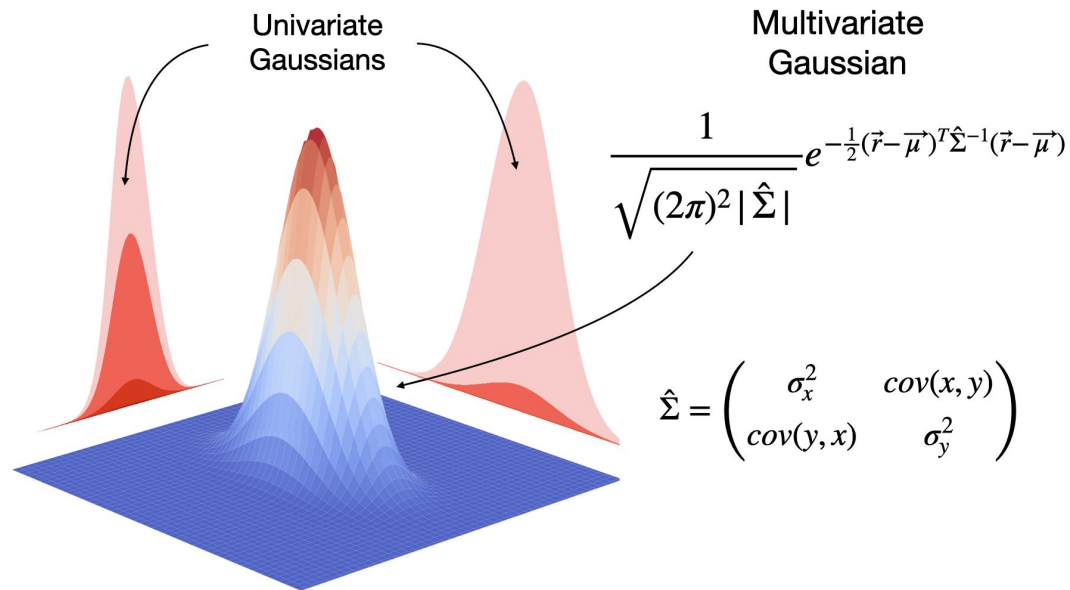
$$c_{NB} = \operatorname{argmax}_{c_j \in C} \log P(c_j) + \sum_{i \in \text{positions}} \log P(x_i | c_j)$$

Underflow Prevention

- Class with highest final un-normalized log probability score is still the most probable.

$$\begin{aligned}\log(P(Y|X_1, \dots, X_n)) &= \log\left(\frac{P(X_1, \dots, X_n|Y) \cdot P(Y)}{P(X_1, \dots, X_n)}\right) \\ &= \text{constant} + \log\left(\prod_{i=1}^n P(X_i|Y)\right) + \log P(Y) \\ &= \text{constant} + \sum_{i=1}^n \log P(X_i|Y) + \log P(Y)\end{aligned}$$

Drawbacks



- Actually, the Naïve Bayes assumption is almost never true
- Still... Naïve Bayes often performs surprisingly well even when its assumptions do not hold

Merits

- Naïve Bayes is:
 - Really easy to implement and often works well
 - Often a good first thing to try
 - Commonly used as a “punching bag” for smarter algorithms

